

Cognitive Integration: Active Inference and Diagrammatic Reasoning

The Missing Layer: A Dynamic Process Theory of Cognition

The preceding sections have developed a rich mathematical infrastructure for analyzing case systems—categorical, type-logical, distributional, enriched, and topos-theoretic. But these frameworks are *static*: they describe the structure of case without explaining how a cognitive agent *uses* that structure in real-time language understanding and production. What is needed is a *process theory* that explains how case-marked relational structure is deployed in the dynamic, embodied, context-sensitive activity of making sense of the world.

Active inference [friston2017active] provides exactly this layer.

Active Inference as a Process Theory of Language Understanding

The Free Energy Principle and Surprise Minimization

Active inference is the process theory derived from the free energy principle (FEP): every self-organizing system maintains itself by minimizing the surprisal (negative log-probability) of its sensory observations under a *generative model* of its environment [-@friston2010free]. The system does this through two complementary strategies:

1. **Perceptual inference:** Update internal beliefs to better predict current observations (reduce prediction error)
2. **Active inference:** Act on the environment to bring observations in line with predictions (reduce expected prediction error)

Recent extensions of active inference to linguistics and cognitive science have modeled language comprehension and production as forms of sequential Bayesian inference. As Donnarumma, Frosolone, and Pezzulo (2023) note in their integration of large language models and active inference, “linguistic processing [is]

Distributional Active Inference: Convergence of Two Distributional Traditions

A remarkable convergence has recently emerged between *distributional semantics* in linguistics and *distributional reinforcement learning* in machine learning, mediated by active inference. Akgül et al. [Akgül2026distributional] introduce **Distributional Active Inference (DAIF)**, which integrates active inference into the distributional RL framework of Bellemare, Dabney, and Munos [Bellemare2017distributional]. Where classical RL optimizes *expected* returns (scalar values), distributional RL models the *full distribution* of returns—a shift from point estimates to distributional representations that parallels the shift from symbolic to distributional semantics in linguistics.

The formal architecture of DAIF proceeds through three stages: (1) reconstructing active inference via variational Bayesian inference on a controlled Markov process, expressing priors through Pearl's do-calculus; (2) defining a *push-forward* operation on representation paths that maps latent-space trajectories to return distributions; (3) defining a *pull-back* operation that maps return distributions back to latent-space trajectories.

CEREBRUM: The Computational Architecture

Architecture and Design Principles

The **CEREBRUM** framework [[@friedman2024cerebrum](#); [@cerebrum2024github](#)][—Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling](#)[—](#)provides a computational architecture that implements the categorical case framework within an active inference engine. CEREBRUM instantiates the view of Vasil et al. [[-@vasil2020world](#)] that human communication is itself active inference: a process of jointly constructing and refining generative models of shared relational structure.

CEREBRUM's key design principles:

Principle	Implementation
Cases as functional roles	Model components carry case markings that determine their computational role in the inference cycle
Morphisms as message	Grammatical relations are

Diagrammatic Reasoning as a Cognitive Process Theory

Diagrams as the Format of Generative Models

The claim from ?? can now be strengthened: commutative diagrams are not merely convenient representations of case structure—they are the *format* in which cognitive agents maintain their generative models of relational structure.

This claim is supported by converging evidence:

1. **Computational advantage** (Larkin & Simon, [-@larkin1987diagram]): Diagrams enable search, recognition, and inference operations that are computationally prohibitive in sentential format. A commutative case diagram allows the agent to verify consistency (does the direct path equal the composed path?) by simple spatial inspection.
2. **Free ride inferences** (Shimojima, [-@shimojima1996reasoning]): Properties of the diagram that are perceptually available but would require explicit computation in a sentential format. In a case diagram, transitivity of grammatical relations is *visible*—the existence of a path from NOM to DAT through ACC is spatially apparent.

Total Cognitive Scenario Understanding: The Integrated Framework

The full picture emerges when we combine all five pillars within the active inference framework:

1. **Case categories** (??) provide the *objects and morphisms* of the generative model—the vocabulary of roles and relations
2. **Categorical grammar** (??) provides the *composition rules*—how roles combine to form structured derivations
3. **DisCoCat/DisCoCirc** (??) provides the *semantic functor*—mapping syntactic structure to distributional meaning
4. **Enriched structure** (??) provides the *precision parameters*—graded weights that control inference
5. **Topos-theoretic bridges** (??) provide *transfer theorems*—ensuring consistency across formalizations

The active inference agent uses this combined structure as a single, integrated generative model. Each scenario it encounters—a sentence heard, a scene observed, an action planned—is

Computational Verification and Results

The framework developed in this paper is not merely theoretical—it is computationally verified through a comprehensive implementation and test suite that exercises every categorical construction discussed above.

System Architecture and Implementation

Implementation Architecture and Categorical Core

The categorical core (`CaseCategory`, `EnrichedCategory`, `AlignmentFunctor`, `NaturalTransformation`) is implemented in Python with set-based object tracking and list-based morphism storage, enforcing categorical axioms at construction time. Five additional modules extend the core: `FluidSFunctor` (context-dependent alignment parameterized by volition), `CaseDiagramBelief` and active inference computations (variational free energy, prediction error, belief updating), `CasePOVM` and quantum case assignment (POVM-based probability via [eq:8-1]), `DitransitiveSentence` (three-argument verb